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Electrochemical Hydrogen Generation from Waste Produced Water Using Graphene- Based Tungsten Trioxide composites

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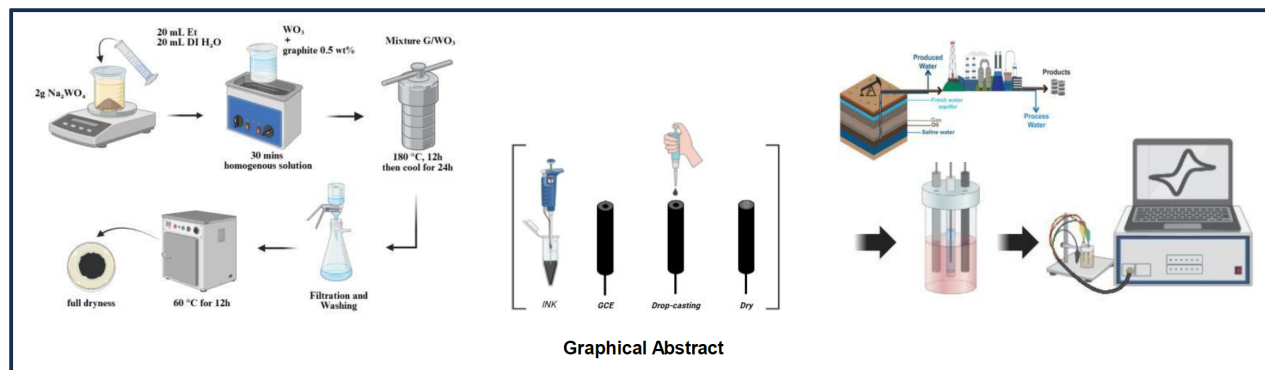
Abstract

This work explores the synthesis and characterization of tungsten trioxide (WO_3) nanocomposites with graphene derivatives, including graphene, graphene oxide (GO), and reduced graphene oxide (rGO), using a hydrothermal method. The composites were prepared by incorporating 0.5 wt.% of each graphene derivative with WO_3 to investigate their impact on hydrogen generation. Structural analysis using X-ray diffraction (XRD), Fourier-transform infrared spectroscopy (FTIR), and Raman spectroscopy confirmed the formation of crystalline WO_3 and the successful incorporation of graphene-based materials. Scanning electron microscopy (SEM) revealed well-integrated composite structures with WO_3 particles anchored on graphene sheets. The electrochemical measurements, including linear sweep voltammetry (LSV), cyclic voltammetry (CV), and electrochemical impedance spectroscopy (EIS), were measured in 0.5 M H_2SO_4 using a three-electrode system to evaluate catalytic activity, charge transfer behaviour, and stability. These results demonstrate the potential of graphene-based WO_3 composites as promising materials for electrochemical hydrogen production, contributing to developing efficient, non-precious metal catalysts.

Introduction

The escalating global need for sustainable energy alternatives has been accelerated by fossil fuel depletion and urgent climate change mitigation requirements, making the investigation of renewable energy sources critically important. Hydrogen stands out as a highly promising solution due to its exceptional energy density and environmentally friendly combustion characteristics, positioning it as a key candidate for addressing contemporary energy challenges^[1,2]. Nevertheless, the development of efficient and economically viable hydrogen production methods continues to pose a major obstacle to its widespread commercial implementation. Electrochemical water splitting, specifically the hydrogen evolution reaction (HER) occurring at the cathode, represents the most viable approach for achieving sustainable hydrogen generation^[3,4]. Although platinum-group metals continue to serve as the gold standard for HER electrocatalysts, exhibiting remarkable activity with overpotentials ranging from 30 to 50 mV at 10

mA cm^{-2} , their commercial deployment faces significant challenges due to excessive costs, natural scarcity, and restricted availability^[5,6]. Therefore, the advancement of abundant, economically feasible alternative catalysts that can deliver comparable electrochemical performance has become crucial for establishing commercially viable hydrogen production processes^[7].



The graphical abstract shows the facile synthesis of graphene-based WO_3 composites electrodes

Tungsten trioxide (WO_3) has attracted considerable attention as an electrode material owing to its superior charge-transport properties, distinctive physicochemical characteristics, and exceptional corrosion resistance^[8,9]. The reversible $\text{W}^{6+}/\text{W}^{5+}$ redox system in WO_3 creates numerous active sites for proton adsorption and subsequent hydrogen generation, while its characteristic layered crystal architecture facilitates efficient ion intercalation processes^[10,11]. Nevertheless, its practical implementation is frequently hindered by relatively limited specific surface area, inadequate electrical conductivity ($\sim 10^{-8} \text{ S cm}^{-1}$), and restricted electrochemically accessible surface area, leading to slow charge transfer dynamics and suboptimal catalytic efficiency^[12,13]. According to recent studies, WO_3 -based catalysts exhibit overpotentials ranging from 200 to 400 mV at 10 mA cm^{-2} , considerably exceeding those of noble metal catalysts, thus requiring strategic enhancement approaches to improve their electrocatalytic effectiveness^[14,15]. The electrochemical reaction mechanism for H^+ ions occurring throughout the surface of the WO_3 electrode can be expressed as follows:



The HER mechanism on WO_3 surfaces involves the intercalation of protons into the tungsten oxide lattice (Equation 4), accompanied by the simultaneous reduction of tungsten centers from higher to lower oxidation states (Equations 5 and 6). This process creates hydrogen bronze phases (H_xWO_3) and generates sub-stoichiometric tungsten oxides with enhanced catalytic activity due to the presence of reduced tungsten species that serve as active sites for hydrogen evolution.

Integrating WO_3 with complementary nanomaterials has been investigated as a strategy to address these constraints and enhance its performance in energy storage and conversion applications. Reduced graphene oxide (rGO), a crystalline carbon material with remarkable properties, exhibits a layered architecture composed of reduced graphene sheets. Individual graphene layers feature carbon atoms organized in hexagonal arrangements, maintained by robust covalent bonds within layers and weak van der Waals interactions between them^[16,17]. rGO demonstrates outstanding electrical conductivity ($>1000 \text{ S m}^{-1}$), extensive specific surface area ($>500 \text{ m}^2 \text{ g}^{-1}$), and superior chemical stability, establishing it as an optimal support material for electrocatalytic processes^[18,19]. Its excellent thermal and electrical conductivity properties make rGO valuable for electrode materials and energy conversion technologies. Contemporary

research emphasizes utilizing rGO and related derivatives to enhance the performance of energy-related technologies, with particular focus on hydrogen evolution applications^[20,21].

The development of heterojunctions between rGO and metal oxides generates synergistic interactions that enhance charge transfer kinetics, expand the availability of active sites, and improve overall catalytic effectiveness^[22,23]. Earlier studies on graphene-metal oxide composites have shown substantial enhancements in HER activity, with optimal carbon content typically falling within 10-30 wt.%^[24,25]. However, comprehensive studies examining ultra-low rGO loadings in WO₃ composites and their influence on HER mechanisms remain scarce. The ideal rGO loading for achieving maximum catalytic efficiency while minimizing material expenses has not been thoroughly investigated, and the fundamental mechanisms controlling hydrogen evolution on WO₃ surfaces, particularly the influence of tungsten oxidation state variations during electrocatalysis, remain inadequately understood^[26,27]. Additionally, the formation and catalytic importance of sub-stoichiometric tungsten oxides during HER have received insufficient attention, despite their potential to function as highly active sites for hydrogen evolution.

Table 1—Comparison of properties and findings from referenced studies on rGO-WO₃ composites for energy applications

Study	Material Used	Application	Key Finding
Balaji et al., 2024	WO ₃ -rGO	Electrochemical & Photocatalysis	Enhanced dual-functionality
Srivastava et al., 2012	rGO-WO ₃	Gas Sensing	Faster catalytic response
Current Study	rGO/WO ₃	HER	Optimized catalytic activity with 0.5 wt.%

This research examines graphene derivatives combined with WO₃ composites as electrocatalysts for hydrogen generation in acidic environments. The composites were prepared using hydrothermal synthesis and analyzed through structural and electrochemical characterization techniques. This work elucidates the function of graphene derivatives in improving charge transfer and catalytic performance while investigating the generation of sub-stoichiometric tungsten oxides during electrocatalytic processes. These results contribute to the advancement of cost-effective, non-precious metal catalysts for sustainable hydrogen production.

Materials and Methods

Materials and Reagents

Graphite powder, Graphene powder, Sodium tungstate dihydrate (Na₂WO₄·2H₂O) ACS reagent, ≥99%, Potassium permanganate (KMnO₄), Hydrogen peroxide 30% (H₂O₂), Nafion from Sigma-Aldrich, (H₂SO₄) sulfuric acid (1.0 M), Potassium Hydroxide (KOH), and Hydrochloric acid (HCl). Distilled water is used throughout the experiments.

Instruments

Using powder X-ray diffraction (PXRD) on a Rigaku MiniFlex diffractometer coupled with Cu-Kα radiation, the crystalline structures of the samples were examined, enabling thorough phase identification and crystallinity evaluation. The morphological characteristics of the samples were studied using the Thermo Scientific Quattro scanning electron microscope (SEM), providing high-resolution images of surface features and particle structures. Fourier-transform infrared (FTIR) spectroscopy was performed using the Thermo Scientific™ Nicolet™ iS10 FTIR spectrometer to investigate chemical bonds and confirm the presence of functional groups. All electrochemical experiments are measured using a CHI 760E electrochemical workstation (CH Instruments Inc., USA).

Synthesis of Graphene/WO₃

2 g of sodium tungstate (Na₂WO₄) dissolved in 40 mL of a solvent mixture consisting of ethanol (Et) and distilled water (DI H₂O). Powdered graphene was added to the solution of 0.5 wt.%. The mixture was sonicated for 30 minutes until a homogeneous dispersion was achieved. The resulting mixtures were then transferred into a Teflon tube in an autoclave and subjected to a temperature of 180 °C for 12 hours. After the autoclave treatment, the samples were removed and cooled for 24 hours. Following the cooling period, filtration was performed using small filter paper to ensure the complete removal of any precipitate. The filtered samples were subsequently placed in an oven at 60 °C for 12 hours to achieve complete dryness.

Synthesis of GO/WO₃

Graphene oxide (GO) was synthesised using a modified Hummers' method. Briefly, 1 g of graphite powder was gradually mixed with 50 mL of concentrated H₂SO₄ and stirred at room temperature for 12 hours. Then, 6 g of KMnO₄ was added slowly over 2.5 hours under continuous stirring. Afterwards, 80 mL of distilled water (DIW) was added, and the mixture was heated to 50 °C until the color shifted from yellowish-brown to dark brown. 200 mL of DIW containing 6 mL of 30% H₂O₂ was added to terminate the reaction. The GO dispersion was purified using dialysis membrane filtration. A cellulose-based dialysis membrane was used to purify the GO by immersion in DI water for 24 hours, during which the water was replaced 4–5 times to remove residual acids and byproducts. After purification, the pH was adjusted to 2–3, and the GO was dried at 80 °C for 24 hours. A 0.5 wt.% dispersion of the prepared GO was then used to synthesize the GO/WO₃ composite using the same method described in Section 2.3.

Preparation of Synthetic Brine to Simulate Produced Water

A synthetic brine solution was prepared to replicate the ionic composition of produced water for electrochemical testing. Pre-weighed amounts of selected salts, as shown in Table 2, were added to distilled water in a volumetric flask. The mixture was stirred continuously for 24 hours to ensure complete dissolution and uniform distribution of all components. The solution was then filtered to remove any undissolved particles or impurities. The resulting clear brine solution was stored and used directly as the electrolyte in subsequent experimental procedures.

Table 2—Composition of Synthetic Brine Used to Simulate Produced Water

Salt	Chemical Formula	Mass
Sodium chloride	NaCl	53.668 g/kg w
Magnesium chloride hexahydrate	MgCl ₂ ·6H ₂ O	8.515 g/kg w
Calcium chloride dihydrate	CaCl ₂ ·2H ₂ O	18.817 g/kg w
Sodium sulfate	Na ₂ SO ₄	1.351 g/kg w

Fabrication of modified electrode

To prepare the catalyst ink, 5.0 mg of the synthesised materials were dispersed in a 2:1 mixture of 200 µL isopropanol (IPA), 100 µL distilled water (DIW), and 30.0 µL Nafion solution (5 wt.%). The suspension was sonicated for 30 minutes to ensure a homogeneous dispersion. A 10.0 µL aliquot of the ink was then drop-cast onto a glassy carbon electrode (GCE) and dried at room temperature, yielding a thin and uniform catalytic layer.

Electrochemical Measurements

The electrocatalytic performance of the G/WO₃-modified electrode was evaluated using a standard three-electrode setup connected to a potentiostat. The prepared glassy carbon electrode was the working electrode, a platinum wire served as the counter electrode, and an Ag/AgCl electrode was used as the reference. Three

electrolytes, 0.5 M H_2SO_4 (sulfuric acid), Synthetic brine (produced water), and a 1:1 mixture of 0.5 M H_2SO_4 and brine, were employed to assess the catalyst's performance. The electrochemical tests conducted included linear sweep voltammetry (LSV), cyclic voltammetry (CV), and electrochemical impedance spectroscopy (EIS). All measurements were conducted under ambient conditions.

Results and Discussion

Structural and Morphological Properties

The morphological features of the synthesised G/ WO_3 nanocomposite, which demonstrates a folded structure of the graphene sheet coupled with WO_3 nanoplates, are shown in Figure 1. The layered sheet structure was beneficial in increasing the surface area of the G/ WO_3 nanocomposite. In addition, the two-dimensional porous structure is expected to facilitate the penetration of the electrolyte and improve the ionic diffusion, which enhances the electrochemical activity.

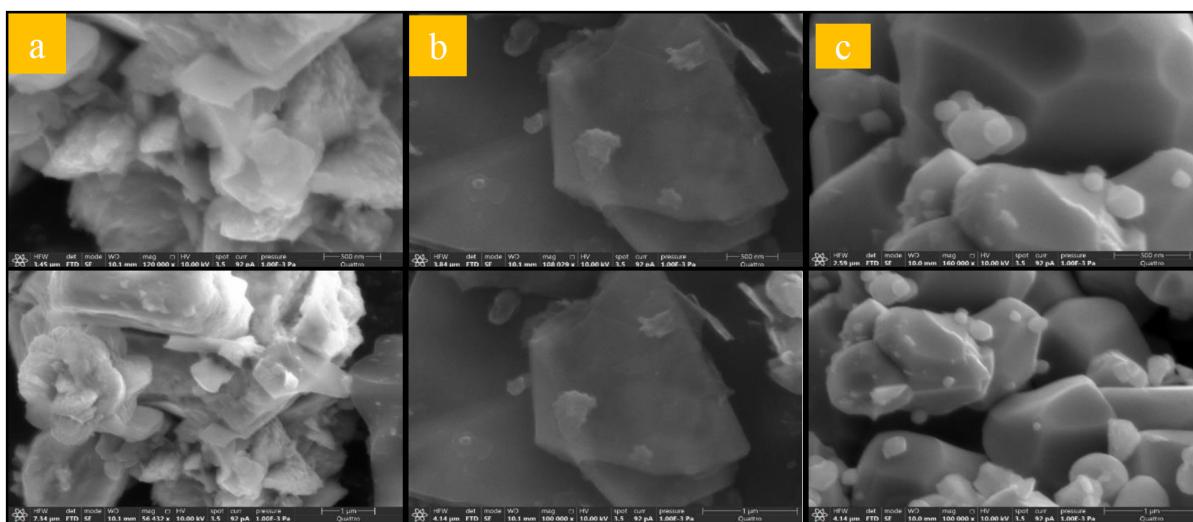


Figure 1—analysis of (a) WO_3 , (b) Graphene and (c) WO_3 Add the bare size

The X-ray diffraction patterns of pure graphene, WO_3 , and the Graphene/ WO_3 composite are shown in Figure 2 (a). The pattern for pure graphene exhibits a sharp and intense peak centered at approximately $2\theta = 26.5^\circ$, corresponding to the (002) crystallographic plane of graphitic carbon. This reflection indicates a high degree of graphitization and layered stacking order, in agreement with previously reported values for crystalline graphite structures.¹ The WO_3 sample displays multiple sharp peaks between $2\theta = 20^\circ$ and 50° , with prominent reflections at approximately 23.1° , 23.5° , 24.3° , 28.2° , and 33.5° , which are attributed to the monoclinic phase of WO_3 and are consistent with JCPDS No. 43-1035 ref. These peaks confirm the formation of a well-crystallized monoclinic WO_3 structure. In the G/ WO_3 composite, the characteristic peaks of WO_3 remain present, but with noticeably reduced intensity and broader profiles. Additionally, the distinct peak of pure graphene at 26.5° is significantly diminished or absent, suggesting successful dispersion of graphene within the WO_3 matrix and potential reduction in stacking order. The overall broadening and decreased intensity in the composite pattern may also indicate reduced crystallite size or partial structural distortion due to the interaction between WO_3 and graphene sheets.

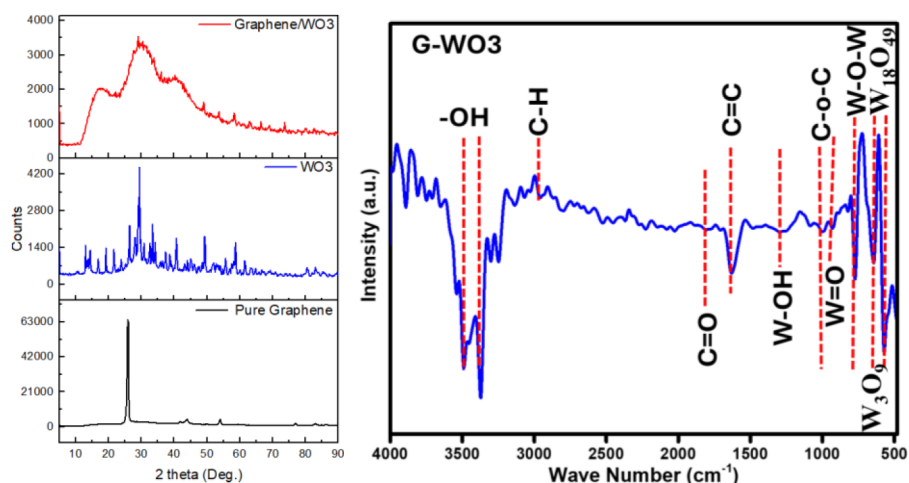


Figure 2—(a) PXRD Patterns and (b) FTIR Spectra Analysis of Pure Graphene, WO_3 and the composite G/WO_3 .

The Fourier-transform infrared (FTIR) spectrum of G/WO_3 composite confirms the successful formation of monoclinic WO_3 structure, as shown in Figure 2 (b). Key characteristic peaks include the terminal $\text{W}=\text{O}$ stretching vibration at 982 cm^{-1} , $\text{W}-\text{O}-\text{W}$ bridging modes at 714 cm^{-1} , and $\text{W}-\text{O}$ stretching vibrations at 620 cm^{-1} . The peak at 832 cm^{-1} corresponds to inter-bridge $\text{O}-\text{W}-\text{O}$ stretching modes, confirming the presence of corner-sharing WO_6 octahedra. The absence of broad absorption around 3380 cm^{-1} indicates effective dehydration during thermal treatment. In comparison, weak features in the $1400\text{--}1700\text{ cm}^{-1}$ region suggest minimal residual oxygen-containing functional groups from the graphene component. These results validate the structural integrity and purity of the synthesized G/WO_3 composite, supporting the XRD findings.

Electrochemical performance

The electrocatalytic performance of the synthesized materials was evaluated using linear sweep voltammetry (LSV) conducted in the potential range of 0 to -1.0 V vs. RHE. Figure 3(a) presents the LSV curves for three samples: G/WO_3 , WO_3 , and the bare glassy carbon electrode (GCE), revealing apparent differences in their catalytic behavior. The bare GCE exhibits negligible electrocatalytic activity across the entire potential window, with current densities remaining near zero even. This silent performance confirms the absence of significant catalytic sites on the unmodified electrode surface. In contrast, both WO_3 and G/WO_3 demonstrate substantially improved electrocatalytic activity. The WO_3 catalyst shows an onset potential of approximately -0.7 V vs. RHE. The G/WO_3 composite exhibits superior HER performance, with a more favorable onset potential of around -0.6 V vs. RHE and a higher current density of 30 mA cm^{-2} , indicating enhanced reaction kinetics and a larger number of active sites.

Electrochemical impedance spectroscopy (EIS) was used to evaluate the charge transfer resistance (R_{ct}) and solution resistance (R_{s}) of the WO_3 , G/WO_3 , and GO/WO_3 composites, as shown in Figure 3(b). Among the tested samples, WO_3 exhibited the highest R_{ct} ($44,488\ \Omega$) and R_{s} ($12.2\ \Omega$), indicating poor charge transfer and high impedance. In contrast, G/WO_3 showed the lowest R_{ct} ($10,659\ \Omega$) and R_{s} ($10.6\ \Omega$), reflecting enhanced charge transfer kinetics and superior conductivity. GO/WO_3 displayed intermediate values, with an R_{ct} of $14,144\ \Omega$ and R_{s} of $16.3\ \Omega$. These results confirm that the incorporation of graphene significantly reduces charge transfer resistance and improves electrochemical performance.

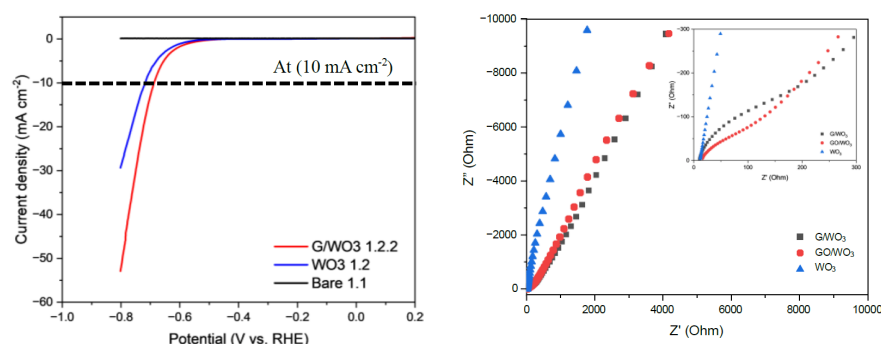


Figure 3—(a) LSV of 0.5% wt. of G/WO₃, WO₃, and GCE bare in acidic media (0.5M H₂SO₄) indicates hydrogen evolution reaction. (b) EIS of the three electrode materials G/WO₃, GO/WO₃ and WO₃ alone

The enhanced performance of the G/WO₃ composite can be attributed to the synergistic interaction between graphene and WO₃, which improves electrical conductivity and provides additional active sites. The graphene component promotes efficient electron transfer and suppresses WO₃ agglomeration, increasing the electrochemically active surface area. Collectively, these findings demonstrate that G/WO₃ is a promising electrocatalyst with improved charge transfer properties and enhanced current response for hydrogen evolution applications.

The cyclic voltammetry (CV) graph illustrates the electrochemical performance of three electrodes: WO₃, GCE (bare glassy carbon electrode), and G/WO₃ (graphene with WO₃ composite). The G/WO₃ composite (blue line) demonstrates the highest current density across the potential range compared to WO₃ (grey line) and GCE (red line), indicating superior electrochemical activity. This enhanced performance is attributed to the synergistic effect of graphene and WO₃, where graphene provides high electrical conductivity, a large surface area, and efficient electron transfer pathways, while WO₃ contributes active catalytic sites. The broad peaks in the G/WO₃ curve suggest efficient redox processes and substantial charge storage, highlighting its potential for HER applications.

The CV studies were initially conducted within a potential range of -1.5 to 0.45 V versus Ag/AgCl. When the composite concentration was adjusted to 0.5 wt.% and the potential range extended to -1.5 to 0.5 V, the data shown in Figure 4 were obtained, effectively covering a range of -1.6 to 0.45 V. During these trials, a notable sharp peak was observed around 0.46 V. This adjustment underscores the importance of optimizing both the composition and potential range to achieve improved electrochemical responses.

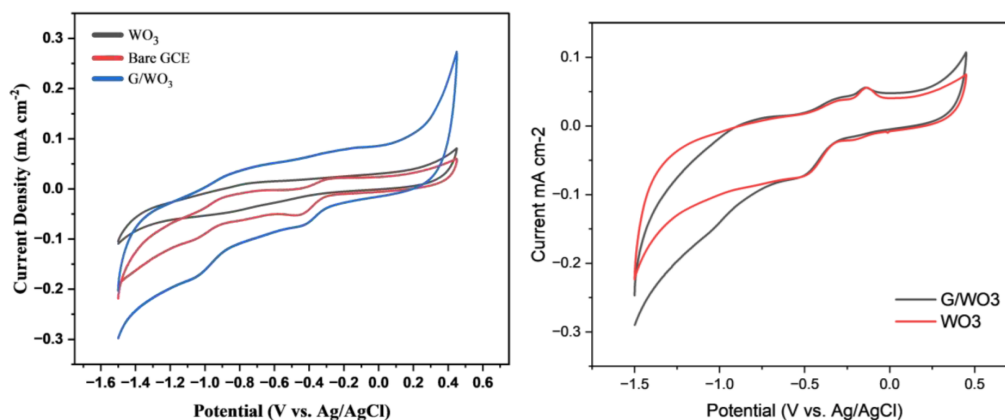


Figure 4—Cyclic voltammetry of (a) 0.5% wt. of G/WO₃, WO₃ and bare GCE at a scan rate of 0.1V and (b) 0.5 Wt.% of G/WO₃ and WO₃.

The cyclic voltammetry data reveal distinct redox behavior for the different electrode materials. In Figure 4 (a), the G/WO₃ electrode exhibits well-defined redox peaks with a prominent oxidation peak around +0.4 V and corresponding reduction peak around −0.2 V vs. Ag/AgCl, which are characteristic of the W⁶⁺/W⁵⁺ redox couple in WO₃. The bare GCE and pure WO₃ electrodes show much smaller redox features, indicating limited electrochemical activity. The G/WO₃ composite demonstrates significantly enhanced current response and better-defined redox peaks compared to its individual components, suggesting improved electron transfer kinetics due to the synergistic effect between graphene and WO₃. In Figure 4 (b), the comparison between G/WO₃ and WO₃ shows that the graphene-WO₃ composite maintains superior electrochemical activity with higher current densities and more reversible redox behavior. The quasi-reversible nature of the redox peaks in G/WO₃ indicates good electrical conductivity and efficient charge transfer.

The linear sweep voltammetry (LSV) curves presented in Figure 5 demonstrate the enhanced electrocatalytic activity of the G/WO₃- modified electrode compared to the bare glassy carbon electrode (GCE) for the hydrogen evolution reaction (HER) in both synthetic brine-based electrolytes. In the produced water (PW) system, Figure 5(a), the G/WO₃ electrode shows significantly improved HER activity, achieving a current density of approximately −0.5 mA cm^{−2} at −1.6 V vs. Ag/AgCl, while the bare GCE exhibits negligible activity across the same potential range. This represents a 5–6-fold enhancement in catalytic performance due to introducing the G/WO₃ composite.

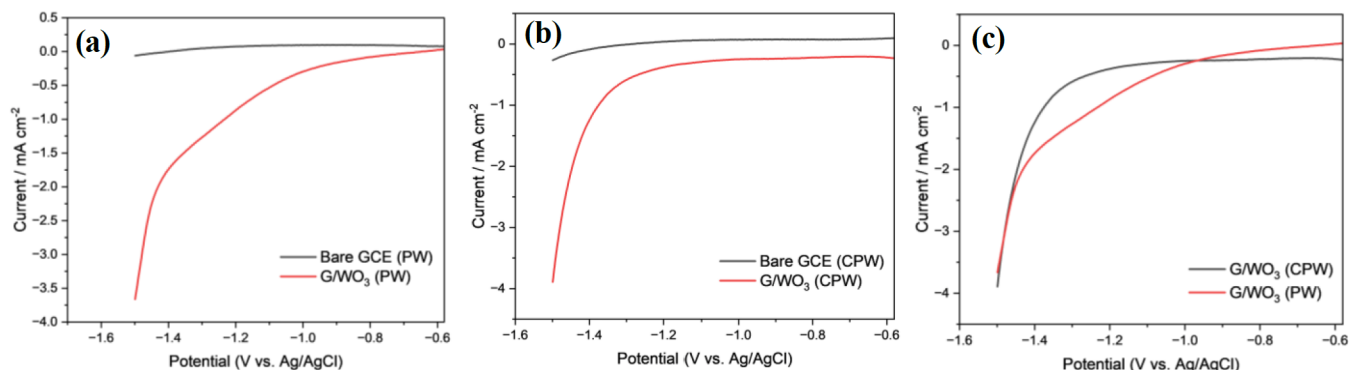


Figure 5—LSV comparing the Bare GCE and G/WO₃ shows the effect of: (a) produced water (PW) and (b) Carbonated produced water (CPW) on HER, respectively. (c) LSV compares the effect of both electrolytes on the same catalyst.

In the carbonated produced water (CPW) system, Figure 5(b), the electrocatalytic activity of the G/WO₃ electrode is further amplified. Saturation of the electrolyte with CO₂ lowers the pH to approximately 4–5, improving proton availability and accelerating HER kinetics. Under these acidic conditions, the G/WO₃ electrode reaches a current density of nearly −4 mA cm^{−2}, compared to the minimal response of the bare GCE, resulting in an 8–10-fold performance increase.

Figure 5(c) directly compares the HER performance of the G/WO₃ electrode in both electrolytes, confirming that carbonation significantly enhances activity. In addition to higher current densities, the G/WO₃ electrode exhibits a more favorable onset potential in both PW and CPW, indicating reduced overpotential requirements for initiating hydrogen evolution. These measurements, performed at a scan rate of 50 mV s^{−1}, highlight the effectiveness of the G/WO₃ composite as a non-precious metal catalyst for hydrogen production in high-salinity environments, with carbonation offering a notable improvement in catalytic efficiency through pH modulation and enhanced ionic conductivity.

Conclusion

This study successfully demonstrated the synthesis, structural characterization, and electrochemical evaluation of graphene-based WO_3 composites for the hydrogen evolution reaction (HER) in acidic and saline environments. Among the synthesized materials, the G/ WO_3 composite exhibited superior electrocatalytic performance, as evidenced by its lower onset potential, higher current density, and significantly reduced charge transfer resistance compared to both pure WO_3 and GO/ WO_3 . The incorporation of graphene enhanced the electrical conductivity, facilitated more efficient electron transfer, and increased the electrochemically active surface area, resulting in improved HER kinetics. Structural analyses confirmed the successful integration of graphene with monoclinic WO_3 , while electrochemical impedance spectroscopy revealed a ~ 4 -fold decrease in R_{ct} for G/ WO_3 relative to WO_3 alone. Furthermore, performance testing in synthetic brine and carbonated brine electrolytes highlighted the additional catalytic benefit provided by CO_2 acidification, with G/ WO_3 achieving an ~ 8 -fold increase in current density in CPW versus PW. These findings underscore the effectiveness of graphene/ WO_3 composites as cost-effective, non-precious metal electrocatalysts for hydrogen production from wastewater-derived electrolytes and suggest their promising applicability in sustainable energy systems.

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